

Distribution of **allelic partition** from neutral selection

m_1 alleles represented ones, m_2 represented twice etc...

$$\mathbb{P}(m_1, \dots, m_n) = \frac{\alpha^\alpha}{\alpha(\alpha + 1) \cdots (\alpha + n - 1)} \prod_{j=1}^n \frac{n!}{j^{m_j} m_j!}$$

with $\sum_{j=1}^n j \cdot m_j = n$

The Sampling Theory of Selectively Neutral Alleles*

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DEDICATED TO THE MEMORY OF KEN KOJIMA



Fisher, Corbet
and Williams
(1943)

Good
(1953)

Good and
Toulmin
(1956)

Ewens
(1972)

Fisher's log-
series model

Prediction of N.
of species in a
further sample

Ewens'
sampling
formula